Married and Never-Married Days: Marriage, Gender and the Full Time Budget in India

Manuscript anonymised for review

October 2026

Abstract

Objective: This study asks which activities differ alongside the higher unpaid work of married Indian women, within the fixed day, and how this varies with co-residence with parents-in-law and spouses' relative resources and ages.

Background: Studies of marriage and time use usually examine one outcome at a time, although differences across activities must sum to zero within 1,440 minutes.

Method: Using all 450,457 diaries of India's Time Use Survey 2024, the study compares 88,337 married and never-married women and 87,364 men aged 20 to 39 across nine activity divisions, with controls. Selection is addressed by restricting ages, reweighting and bias adjustment; 106,858 linked couples give within-household comparisons.

Results: Married women reported 280 minutes more of unpaid domestic and care work per day than never-married women; for men the difference was 26 minutes more. The women's difference was 233 minutes after reweighting and 264 after bias adjustment. Wives did 367 minutes more of unpaid work than their husbands, and their share of the couple's unpaid work was 91%. Co-residence with parents-in-law did not change wives' unpaid work detectably.

Conclusion: Married and never-married women's days differ far more than men's, and a full time-budget accounting shows which activities differ alongside unpaid work.

Keywords: marriage, time use, unpaid work, gender, patrilocality, India

Introduction

Marriage is closely tied to how time is used. In diary studies from high-income countries, married women do more housework than women in other living arrangements, and the gender gap in housework is largest among the married (Shelton & John, 1996; South & Spitze, 1994). Men's housework, in contrast, falls or stays flat when they enter a union (Baxter et al., 2008; Gupta, 1999). Explanations range from specialisation within the household (Becker, 1985, 1991) and bargaining over the division of labour (Lundberg & Pollak, 1993; Manser & Brown, 1980) to the performance of gender in everyday interaction (Akerlof & Kranton, 2000; Brines, 1994; West & Zimmerman, 1987).

India offers a sharp setting for these questions. In the Time Use Survey 2024, 99% of women and 95% of men aged 35 to 39 have ever been married. Residence after marriage is usually with the husband's family (Dyson & Moore, 1983; Khalil & Mookerjee, 2019): 29% of married women aged 20 to 39 in the survey are daughters-in-law of the household head. Married women's employment is low and has not risen with education (Afridi et al., 2018; Chatterjee et al., 2018; Klasen & Pieters, 2015). For most Indian women, marriage is the transition that sets the terms of adult time use.

Most studies of marriage and time use take one outcome at a time. Separate literatures examine housework (Coltrane, 2000; Lachance-Grzela & Bouchard, 2010), sleep (Burgard & Ailshire, 2013; Maume et al., 2010), free time (Mattingly & Bianchi, 2003), exercise (Nomaguchi & Bianchi, 2004) and social ties (Sarkisian & Gerstel, 2016). A day is a fixed budget, however. Any increase in unpaid work must come out of paid work, learning, leisure, social life, self-care or sleep. Looking at one outcome at a time cannot tell which, and a set of

separate "marriage penalties" estimated on different samples and specifications need not add up. For India, Janiso et al. (2024) use the Time Use Survey 2019 to decompose the large gap between women and men in unpaid housework and care work; this paper instead compares married with never-married women and men, across the whole day, using the 2024 round. This paper estimates the components together. I decompose the married–never-married difference into all nine divisions of the International Classification of Activities for Time-Use Statistics 2016 (ICATUS 2016, United Nations Statistics Division, 2016). The divisions partition the day, so the components sum to zero by construction, and sub-rows for sleep, socialising, exercise and religious practice show where these commonly studied outcomes sit within the budget.

The paper makes three further contributions. First, it treats selection into marriage as a central problem rather than a caveat. Never-married women in their thirties are rare in India, and they are likely to differ from married women in schooling, work orientation and health (Goldman, 1993). I report the decomposition for women aged 20 to 26, where never-married women are least selected; reweight the never-married to the covariate distribution of the married (DiNardo et al., 1996); compute bias-adjusted coefficients and breakdown values in the spirit of Oster (2019); and trace how the gap changes with age as the never-married become more selected. Second, it uses household structure to examine patrilocality directly. Comparing daughters-in-law of the head with wives who head their own household (or whose husband does) separates two accounts of the joint family: one in which the mother-in-law shares domestic work, and one in which the junior wife carries the heaviest load (Kandiyoti, 1988). Third, it links spouses' diaries within households and examines the within-couple

division of unpaid work by the wife's employment, spouses' relative education and the spousal age gap. These comparisons hold the household constant and do not depend on the never-married comparison group.

Married women aged 20 to 39 report 279.7 minutes (SE 2.5) more unpaid domestic and care work per day than never-married women of the same age, education, caste, religion and state. The corresponding difference for men is 26.2 minutes (SE 0.9) more, and the gendered difference in a pooled model is 261.9 minutes (SE 2.6). For women, the largest offsetting reductions are in learning (119 minutes) and employment (incl. travel) (85 minutes). Married women slept 9 minutes less than never-married women. Among linked spouses, wives' unpaid work exceeds husbands' by 366.9 minutes (SE 1.1) a day, and husbands' paid work exceeds wives' by 336.4 minutes (SE 1.5).

Background

Marriage and the division of household labour

The finding that marriage widens the gender gap in housework is among the most replicated in family research. South & Spitze (1994) show that women in married-couple households spend more time on housework than women in other living arrangements. Panel studies find that men's housework falls when they marry or cohabit (Gupta, 1999), and that women's housework responds more than men's to union and parenthood transitions (Baxter et al., 2008; Sanchez & Thomson, 1997). Trend studies document a narrowing of husbands' and wives' housework since the 1960s, but no convergence to equality (Bianchi et al., 2000; Kan et

al., 2011; Sayer, 2005). Cross-national studies show that the gap is larger where macro-level gender inequality is greater and where policies support a male-breadwinner arrangement (Fuwa, 2004; Hook, 2006, 2010). Reviews of this literature agree on the main correlates (relative resources, time availability and gender ideology) but not on their relative weight (Coltrane, 2000; Davis & Greenstein, 2009; Lachance-Grzela & Bouchard, 2010; Shelton & John, 1996).

Two families of theory organise this work. The first is economic. In Becker's model of the household, spouses specialise according to comparative advantage, and small initial differences in market productivity, reinforced by investment in sex-specific human capital, lead to complete specialisation (Becker, 1985, 1991). Home production competes with both market work and leisure (Becker, 1965; Gronau, 1977). Bargaining models replace the unitary household with spouses whose outcomes depend on their threat points: divorce in Manser & Brown (1980) and McElroy & Horney (1981), a non-cooperative marriage governed by traditional roles in the "separate spheres" model of Lundberg & Pollak (1993), and a general collective rule in Chiappori (1992; see also Lundberg & Pollak, 1996). In bargaining models, a spouse's earnings, education and outside options buy relief from housework. Sociologists reach a similar prediction through relative-resources theory (Blood & Wolfe, 1960). Feminist economists have broadened the bargaining framework to include social norms and the extra-household institutions that determine what women can claim (Agarwal, 1997; Sen, 1990).

The second family treats housework as a way of performing gender. In the "doing gender" perspective, men and women accomplish their gender in everyday interaction, and the division of housework is a central arena for doing so (West & Zimmerman, 1987). Brines (1994) found

that economically dependent husbands did less housework, not more, as resource theory would predict. Later work called this pattern gender display or deviance neutralisation (Evertsson & Nermo, 2004; Greenstein, 2000). Bittman et al. (2003) found that wives' housework falls as their share of earnings rises toward equality, but that in some settings the relationship reverses once wives out-earn their husbands. Akerlof & Kranton (2000) formalised the idea in economics: people lose utility when they act against the prescriptions of their gender category. Bertrand et al. (2015) find that American wives who out-earn their husbands do more home production than their earnings would predict. The evidence is contested. Gupta (2007) argues that women's absolute earnings, not their earnings relative to their husbands', predict their housework. Killewald & Gough (2010) show that the relationship with wives' earnings is strongly non-linear, and Schneider (2011) finds evidence of gender performance for both husbands and wives. Structural accounts locate gender at several levels at once, in individual identities, in interaction and in institutions (Ridgeway & Correll, 2004; Risman, 2004), which is one reason couple-level resource effects may be weak where institutional constraints are strong.

The time budget as a whole

Time-budget research has long argued that activities should be studied together because the day is fixed (Gershuny, 2000; Juster & Stafford, 1991; Robinson & Godbey, 1997). Aguiar & Hurst (2007) and Gimenez-Nadal & Sevilla (2012) use complete time budgets to trace substitution between market work, home production and leisure. Burda et al. (2013) show that in rich countries men and women do roughly the same total amount of paid and unpaid

work, while in poorer countries women do more. Marriage could therefore raise women's total work, or it could shift work from the market to the home with little change in total work. Only a full accounting distinguishes the two.

Free time differs in quality as well as quantity. Women's leisure is more often fragmented or combined with care (Bittman & Wajcman, 2000; Mattingly & Bianchi, 2003). Marriage may also reshape social life. Coser (1974) described institutions that claim undivided commitment as "greedy", and Sarkisian & Gerstel (2016) find that married Americans have fewer ties to kin, friends and neighbours than the never-married. In this view, social time, not paid work, is what marriage takes away.

Sleep, health behaviour and marriage

Sleep is the largest single use of time, and it is not fixed. Biddle & Hamermesh (1990) show that sleep responds to the value of time, and in the American Time Use Survey work time is the activity most strongly traded off against sleep (Basner et al., 2007). Gender matters for sleep as well. Women's sleep is more often interrupted by caregiving (Burgard & Ailshire, 2013), and qualitative studies describe a "fourth shift" of night-time care that mothers rather than fathers carry (Maume et al., 2010; Venn et al., 2008). Self-reported sleep problems follow both gender and socio-economic position (Arber et al., 2009). If marriage adds morning and evening domestic tasks (cooking before others wake, cleaning up after others eat), sleep is a plausible margin of adjustment.

Marriage is also associated with health behaviour. Spouses monitor and regulate each other's health behaviour (Umberson, 1987, 1992), and the married have better health and longer

lives in many settings (Ross et al., 1990; Waite, 1995). How much of this is a causal effect of marriage and how much is selection of the healthy into marriage is a long-standing problem (Goldman, 1993). Time-diary evidence on exercise shows that marriage, parenthood and employment relate differently to men's and women's exercise time (Nomaguchi & Bianchi, 2004). In a time-budget framework, sleep, exercise and personal care are components of division 9 (self-care) and division 8 (leisure and sport), so their marriage gaps can be read against the gaps in work.

Marriage in India

Three features of Indian marriage bear on time use. The first is patrilocality. A woman usually moves into her husband's household at marriage, and in many regions marries outside her natal village (Dyson & Moore, 1983; Rahman & Rao, 2004). In a joint household, the new wife is junior to her mother-in-law. Kandiyoti (1988) describes a "patriarchal bargain" in which young brides accept subordination and heavy work in return for authority later in life, when they become mothers-in-law. Recent studies find that co-residence with in-laws is associated with less say over household decisions and less freedom of movement (Debnath, 2015; Khalil & Mookerjee, 2019), and that mothers-in-law restrict daughters-in-law's mobility and social contacts outside the household (Anukriti et al., 2020). Regional differences in women's autonomy are large and are not explained by religion alone (Jejeebhoy & Sathar, 2001). Joint residence could reduce a young wife's domestic load if senior women share the cooking and cleaning, or increase it if the junior wife does the heaviest work. Its effect is an empirical question that the household roster can address.

The second is the timing and form of marriage. Most marriages are arranged by families, though women's role in choosing their husbands has grown (Allendorf & Pandian, 2016). Age at marriage reflects the gender scripts of the community (Desai & Andrist, 2010). Labour-market opportunity can delay marriage (Jensen, 2012), and in Bangladesh later marriage raises girls' schooling (Field & Ambrus, 2008). Because marriage timing reflects education, family resources and work prospects, the women who remain unmarried in their late twenties and thirties are a selected group. This is the main threat to any comparison of married and never-married women.

The third is women's employment. Women's labour force participation in India is low by international standards and fell for much of the 2000s and 2010s, with a particularly marked decline among married women in rural areas (Afridi et al., 2018; Fletcher et al., 2017; Mehrotra & Parida, 2017). Proposed explanations include rising household incomes, the structure of labour demand, the status costs of women's market work, and social norms about women's mobility and work outside the home (Eswaran et al., 2013; Field et al., 2021; Jayachandran, 2021; Klasen & Pieters, 2015). Time-use data show that Indian women carry most of the unpaid work and much of the unrecorded production for own use (Hirway & Jose, 2011). Family structure also shapes women's employment (Dhanaraj & Mahambare, 2019). If marriage moves women out of paid work and into domestic work, the time budget will show it directly. If instead marriage adds domestic work on top of existing activities, leisure, social time or sleep must absorb it.

Hypotheses

The theories above make overlapping predictions about the size of the marriage gap and distinct predictions about its sources and moderators. I state them as expectations that the data can contradict.

H1 (gendered reallocation). Marriage is associated with a larger increase in unpaid domestic and care work for women than for men. Specialisation, bargaining with gendered threat points, doing gender and patrilocal norms all predict this.

H2 (sources of the additional unpaid time). Specialisation predicts that women's additional unpaid work is financed mainly by lower paid work and learning, so total work changes little (H2a). A time-squeeze or greedy-institution account predicts that it is financed mainly by lower leisure, social time and self-care, including sleep, so total work rises (H2b). Because sleep responds to the value of time but has a biological floor (Basner et al., 2007; Biddle & Hamermesh, 1990), I expect sleep to adjust less than free time.

H3 (patrilocal co-residence). If senior women share domestic work, daughters-in-law of the head do less unpaid work than wives in households without a parent generation (H3a, substitution). If the household hierarchy assigns the heaviest work to the junior wife and restricts her movement (Anukriti et al., 2020; Kandiyoti, 1988), daughters-in-law do more unpaid work, less paid work and less social time outside the home (H3b, hierarchy).

H4 (relative resources within couples). Bargaining and relative-resources theories predict that the wife–husband gap in unpaid work is smaller when the wife is employed, when she is more educated than her husband, and when the spousal age gap is smaller. Age gaps proxy

for seniority in a hierarchical marriage and are larger when women marry young (Desai & Andrist, 2010). Doing-gender accounts predict weaker, or non-monotonic, associations with the wife's relative position (Brines, 1994; Greenstein, 2000).

Method

Data

I use the 2024 round because it is the most recent and its diaries cover every household member aged six and above; the 2019 round (National Statistical Office, 2020) uses the same design, and a 2019–2024 comparison of the headline married–never-married difference is planned as an extension. The Time Use Survey 2024, conducted by India's National Statistics Office from January to December 2024, records the activities of every household member aged six and above over the 24 hours from 4:00 a.m. on the day before the interview (National Statistics Office, 2025). Activities are coded to the three-digit level of ICATUS 2016 (United Nations Statistics Division, 2016). Where activities overlap, the survey flags one as the major activity. I use every activity record (10,211,478 records from 450,457 diaries in 139,487 households) and aggregate major activities to the person-day, so all but 5 of the 450,457 diaries partition exactly 1,440 minutes. The roster lists 454,192 persons aged six and above; 3,735 of them have no diary records in the file, which leaves the 450,457 diaries, and the 49 diaries of third-gender respondents are excluded from the comparisons of women and men. Some diaries are reported by another household member; I control for proxy reporting and drop proxy diaries in a robustness check. All estimates use the survey multipliers, and

standard errors are clustered on the first-stage unit, the primary sampling unit (Abadie et al., 2023; Solon et al., 2015).

Samples

The main sample consists of never-married and currently married women and men aged 20 to 39 (88,337 women and 87,364 men). Widowed, divorced and separated persons are excluded from the main comparison and studied separately at ages 50 to 79. In the main age range, 85% of women and 65% of men are currently married.

The couple sample uses the household roster to link spouses. A head is linked to the spouse of the head, and a married child of the head is linked to the child's spouse when each is unique in the household and the two are of opposite sex. 89% of married persons in the survey have a linked spouse diary. Couples with several married children of the head in the same household are not linked, which under-represents the largest joint households. I keep linked wives aged 20 to 59 (106,858 couples; 106,424 have complete covariates and enter the couple regressions). Household members' diaries are collected in the same visit; in 100% of linked couples the two diaries refer to the same day of the week.

Measures

The outcomes are minutes per day in the nine ICATUS divisions: employment and related activities (including work travel), production of goods for own final use (for example farming for the household, fetching water and collecting firewood), unpaid domestic services, unpaid caregiving for household members, volunteering, learning, socialising and religious practice,

culture and leisure (including mass media and sport), and self-care and maintenance (including sleep, eating and personal care). Unpaid work is the sum of domestic services and caregiving, the time-use measure behind Sustainable Development Goal indicator 5.4.1. Sub-rows report sleep, socialising (talking and getting together), exercise and sport, religious practice and mass media.

Marital status and relationship to the household head come from the roster. For married women, I distinguish three residence types: (a) the woman is the head or the head's wife and no parent or parent-in-law of the head lives in the household ("no parent generation"), (b) she is the head or the head's wife and a parent or parent-in-law of the head is present, and (c) she is the daughter-in-law of the head (roster code 4). Among married women aged 20 to 39, these groups account for 60%, 8% and 29% respectively; the remaining 3% hold other positions (for example, a married daughter of the head). For men, the junior category is a married son of the head (code 3). Type (c) is the clearest indicator of patrilocal co-residence. Type (b) mixes households in which the parent generation is the husband's and households in which it is the wife's, because the roster records the relation to the head only.

Covariates are single-year age, education (four tiers), urban residence, caste group (Scheduled Tribe, Scheduled Caste, Other Backward Class, other), religion, state, day of the week and proxy reporting. The baseline model does not control for household size, older household members or consumption. These are partly consequences of marriage, since at marriage a woman usually moves into a different and often larger household, and controlling for them would remove part of the association I want to measure. They enter an extended specification as a robustness check.

Analytic strategy

For each activity division k, separately for women and men,

$$y_i^k = \beta^k \,\text{Married}_i + X_i'\gamma^k + \alpha_{a(i)} + \mu_{s(i)} + \tau_{d(i)} + \varepsilon_i^k,$$

where α, μ and τ are single-year age, state and day-of-week fixed effects. Because the nine divisions sum to 1,440 minutes for all but 5 of the 450,457 diaries and every regression uses the same sample, weights and regressors, $\sum_k \hat{\beta}^k = 0$ up to rounding. The coefficients decompose the adjusted married–never-married difference into its sources. A pooled model with a married × woman interaction and sex-specific age, state and day effects tests whether the unpaid-work difference is gendered. Heterogeneity by residence replaces the married indicator with indicators for the three residence types, each compared with the same never-married group. A model restricted to married women tests whether daughters-in-law differ from wives in households without a parent generation.

The coefficient $\hat{\beta}^k$ is not the causal effect of marriage. The survey is a single cross-section, it does not record the date of marriage, and the never-married are a selected and shrinking group. I use four strategies to judge how much selection could matter. (1) I re-estimate the decomposition for women aged 20 to 26, where 32% of women are never married and the comparison group is closest to the population. (2) I reweight never-married respondents to the joint distribution of the covariates among the married (DiNardo et al., 1996). The weights are the odds of being married from a logit on single-year age, state, education, urban residence, caste and religion (Imbens, 2004; Rosenbaum & Rubin, 1983), with propensities

capped at 0.99. The reweighted regression, which also includes the controls, estimates the average difference among the married, conditional on observables, and is doubly robust to misspecification of either the propensity or the outcome model. (3) For women, I compute bias-adjusted coefficients that assume selection on unobservables is as strong as selection on observables ($\delta = 1$) and that the maximum attainable R^2 is 1.3 times the controlled R^2, and report the value of δ at which the adjusted coefficient would reach zero (Altonji et al., 2005; Oster, 2019). (4) I estimate the unpaid-work gap separately in five age bands. If selection of the never-married drives the gap, the gap should grow sharply as the never-married become rarer.

Within-couple comparisons difference out everything shared by the spouses: household composition, consumption, location, the diary day and much of the social environment. I report the wife-minus-husband gap in unpaid work, the wife's share of the couple's unpaid work, and the husband-minus-wife gap in employment time, by the wife's employment, relative education, residence type and the spousal age gap. Adjusted associations for the age gap and residence type control for the wife's single-year age, both spouses' education, caste, religion, urban residence, state and day of the week. These comparisons describe the division of labour within marriage, not the effect of marriage, and the wife's employment is itself a joint decision of the couple and household.

Table 1: Weighted means by sex and marital status, ages 20–39

	Women, never married	Women, married	Men, never married	Men, married
Employment (incl. travel), min	120	70	323	452
Own-use production of goods, min	12	23	10	19
Unpaid domestic services, min	164	356	29	26
Unpaid caregiving, min	5	103	2	35
Learning, min	154	2	92	2
Socialising, community, religion, min	91	90	96	87
Culture, leisure, media, sport, min	169	112	179	118
Self-care incl. sleep, min	723	684	709	699
of which: sleep, min	544	527	527	515
Age, years	23.4	29.8	24.8	31.7
Employed (usual status)	28%	22%	68%	98%
Current student	32%	0%	19%	0%
Graduate	45%	16%	32%	20%
Urban	43%	29%	41%	31%
Diaries (N)	13,940	74,397	31,598	55,766

Notes: Minutes per day from major activities; the nine divisions (eight shown plus volunteering) sum to 1,440. Weighted by survey multipliers.

Results

Descriptive patterns

Table 1 describes the main sample. Never-married women in this age range are much more likely to be students (32%, against 0% of married women), and 28% of never-married women report being employed in their usual principal status, against 22% of married women. The raw differences between columns mix the association with marriage and the different age and education profiles of the groups, which the regressions below hold constant.

Table 2: Married minus never-married, minutes per day, ages 20–39

Activity	Women	Men
Employment (incl. travel)	-84.7*** (3.3)	69.4*** (2.7)
Own-use production of goods	-1.0 (0.7)	2.4*** (0.6)
Unpaid domestic services	164.8*** (2.3)	-5.5*** (0.8)
Unpaid caregiving	114.9*** (1.1)	31.7*** (0.5)
Volunteering	-0.2 (0.2)	-0.8*** (0.2)
Learning	-118.7*** (2.6)	-36.1*** (1.2)
Socialising, community, religion	-6.6*** (1.4)	-11.0*** (1.2)
Culture, leisure, media, sport	-41.3*** (1.7)	-42.2*** (1.3)
Self-care incl. sleep	-27.1*** (1.6)	-7.9*** (1.3)
Sum of the nine divisions	0.0	-0.0
of which: sleep	-8.6*** (1.4)	-8.7*** (1.1)
of which: socialising	-8.8*** (1.1)	-11.8*** (1.0)
of which: exercise and sport	-2.2*** (0.2)	-3.5*** (0.3)
of which: religious practice	2.7*** (0.5)	0.5 (0.4)
of which: mass media	-35.1*** (1.4)	-25.8*** (1.0)
Unpaid work (domestic + care)	279.7*** (2.5)	26.2*** (0.9)

Notes: Coefficients on currently married from separate regressions by sex, with single-year age, state and day-of-week fixed effects and controls for education, urban residence, caste, religion and proxy report. FSU-clustered standard errors in parentheses. The nine divisions partition the day, so their coefficients sum to zero. * p<0.10, ** p<0.05, *** p<0.01.

Decomposing the married–never-married difference

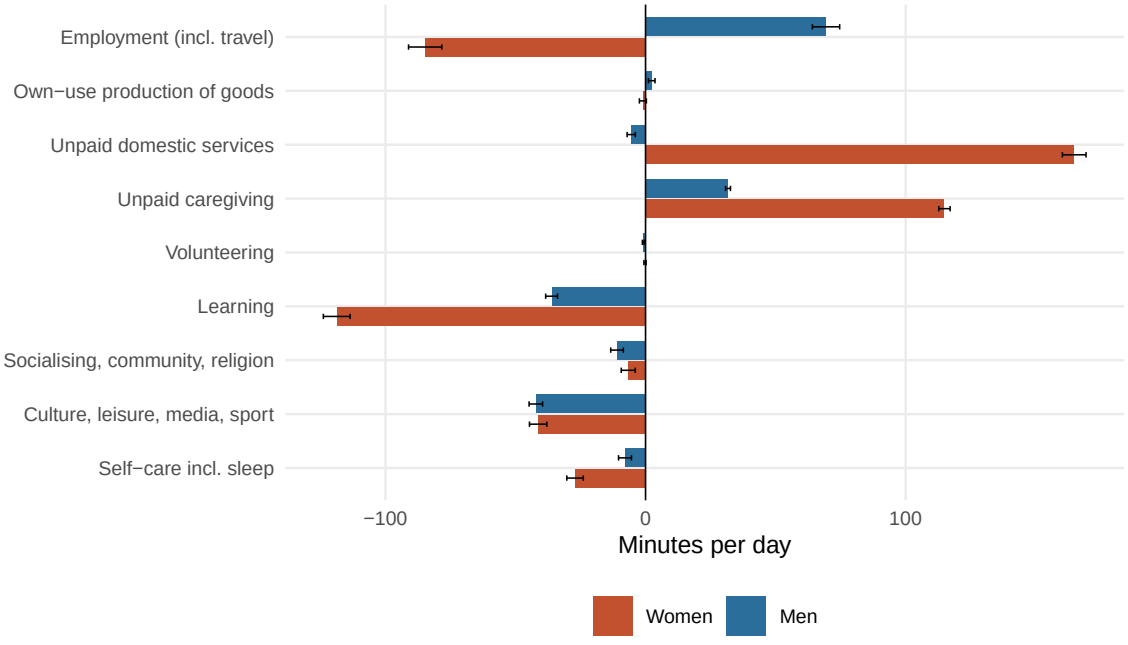


Figure 1: Married minus never-married minutes per day by activity division, ages 20–39. Bars for each sex sum to zero; lines are 95% confidence intervals.

Table 2 and Figure 1 report the decomposition. Marriage is associated with 164.8 minutes (SE 2.3) more unpaid domestic work and 114.9 minutes (SE 1.1) more caregiving for women. For men the corresponding coefficients are -5.5 minutes (SE 0.8) and 31.7 minutes (SE 0.5). In the pooled model, the married × woman coefficient on unpaid work is 261.9 minutes (SE 2.6) (statistically significant at the 1% level). The pattern supports H1: marriage is associated with much more unpaid work for women than for men.

Part of this difference reflects motherhood rather than marriage as such: married women are far more likely to live with young children, and caregiving accounts for 115 of the 280 minutes. Among women in households with no child under six (38,541 women, 70% of them married), the married–never-married difference in unpaid work is 186.6 minutes (SE 3.0): 175.5 minutes (SE 2.9) in domestic work and 11.1 minutes (SE 0.5) in caregiving. The domestic-work difference largely persists without young children in the household, whereas most of the caregiving difference does not. The estimates are therefore best read as the difference that accompanies marriage, including the motherhood that usually follows it, not as what marriage alone adds.

Among the other seven divisions, women's additional unpaid minutes are offset mainly by lower time in learning (119 minutes), employment (incl. travel) (85 minutes) and culture, leisure, media, sport (41 minutes). Employment and learning together account for 73% of the reductions, socialising and leisure for 17%, and self-care for 10%. Most of the offset comes from paid work and learning, which is closer to the specialisation account (H2a) than to a squeeze on free time.

The sub-rows place the commonly studied "marriage penalties" within the budget. Married

women slept 9 minutes less than never-married women (SE 1.4). The coefficients for socialising, exercise and sport, and religious practice are -8.8 minutes (SE 1.1), -2.2 minutes (SE 0.2) and 2.7 minutes (SE 0.5). Sleep moves less than free time, as H2 anticipated. For men, the largest coefficient in absolute value is on employment (incl. travel), at 69.4 minutes (SE 2.7).

Patrilocal co-residence

Table 3: Married minus never-married by residence type, minutes per day, ages 20–39

	Women			Men		
Activity	No parent gen.	Parent gen.	Daughter-in-law	No parent gen.	Parent gen.	Married son
Employment	-82.4*** (3.5)	-74.3*** (4.1)	-91.6*** (3.3)	75.5*** (2.9)	66.6*** (4.5)	65.8*** (2.9)
Own-use goods	3.1*** (0.8)	2.0* (1.1)	-4.3*** (0.7)	4.0*** (0.8)	4.3*** (1.5)	0.5 (0.7)
Domestic services	166.1*** (2.5)	176.7*** (3.1)	170.2*** (2.4)	-5.5*** (0.9)	-8.6*** (1.3)	-8.2*** (0.8)
Caregiving	118.1*** (1.3)	108.8*** (1.7)	113.6*** (1.3)	37.6*** (0.6)	28.3*** (1.0)	25.9*** (0.6)
Volunteering	-0.1 (0.2)	-0.1 (0.4)	-0.4* (0.2)	-0.8*** (0.2)	-0.9*** (0.3)	-1.0*** (0.2)
Learning	-114.9*** (2.6)	-116.1*** (2.7)	-122.3*** (2.6)	-33.5*** (1.2)	-33.4*** (1.3)	-39.0*** (1.2)
Social, community, religion	-7.5*** (1.5)	-10.8*** (2.0)	-6.2*** (1.4)	-16.2*** (1.4)	-5.6** (2.3)	-6.2*** (1.4)
Leisure, media, sport	-48.8*** (1.9)	-49.6*** (2.2)	-35.8*** (1.7)	-49.0*** (1.5)	-41.4*** (2.2)	-34.7*** (1.4)
Self-care	-33.6*** (1.8)	-36.6*** (2.3)	-23.0*** (1.7)	-12.2*** (1.5)	-9.4*** (2.1)	-3.1** (1.3)
of which: sleep	-9.5*** (1.6)	-15.6*** (2.0)	-8.1*** (1.5)	-11.3*** (1.3)	-9.5*** (1.9)	-5.8*** (1.2)
Unpaid work	284.2*** (2.7)	285.5*** (3.4)	283.7*** (2.6)	32.1*** (1.1)	19.7*** (1.6)	17.7*** (1.0)

Notes: Each married residence type is compared with never-married persons of the same sex. "No parent gen.": head or head's spouse, no parent or parent-in-law of the head in the household. "Parent gen.": head or head's spouse with a parent or parent-in-law of the head present. "Daughter-in-law": roster code 4 (women); "Married son": roster code 3 (men). Controls and fixed effects as in the baseline decomposition. FSU-clustered SEs in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table 3 splits married women and men by residence type. Each married group is compared with never-married persons of the same sex, so each column again sums to zero across divisions. The comparison of interest is between daughters-in-law of the head and wives in households without a parent generation. In a model restricted to married women, daughters-in-law do 2.1 minutes (SE 1.8) less unpaid work (not statistically distinguishable from zero), 12.6 minutes (SE 1.7) less employment, 0.8 minutes (SE 1.1) more socialising, community and religious activity, and sleep 1.1 minutes (SE 1.1) more. Unpaid work does not differ detectably by residence type, so the data support neither the substitution account (H3a) nor the hierarchy account (H3b) as a description of average unpaid time.

Two cautions apply. Daughters-in-law are younger on average and at an earlier stage of marriage than wives who head their own household. The single-year age effects absorb the age difference, but not marital duration, which the survey does not record. In addition, the move from a joint to a nuclear household is itself selective, and can follow the death of the senior couple or household partition (Debnath, 2015).

Selection into marriage

Table 4: Women: married minus never-married under alternative treatments of selection, minutes per day

Activity	Baseline	Ages 20–26	Reweighted	Bias-adjusted	Breakdown delta
Employment (incl. travel)	-84.7*** (3.3)	-68.3*** (3.2)	-72.0*** (6.1)	-96.2	-7.40
Own-use production of goods	-1.0 (0.7)	0.3 (0.8)	3.4* (1.8)		
Unpaid domestic services	164.8*** (2.3)	161.6*** (2.6)	141.6*** (5.3)		
Unpaid caregiving	114.9*** (1.1)	105.4*** (1.4)	91.4*** (1.7)		
Volunteering	-0.2 (0.2)	-0.0 (0.3)	-0.1 (0.3)		
Learning	-118.7*** (2.6)	-131.2*** (3.0)	-30.7*** (1.5)	-43.0	1.57
Socialising, community, religion	-6.6*** (1.4)	-4.4*** (1.7)	-18.7*** (2.9)	-8.2	-4.30
Culture, leisure, media, sport	-41.3*** (1.7)	-38.0*** (2.0)	-67.9*** (3.8)	-33.1	5.04
Self-care incl. sleep	-27.1*** (1.6)	-25.3*** (1.9)	-47.1*** (3.4)		
of which: sleep	-8.6*** (1.4)	-8.9*** (1.6)	-26.9*** (2.8)	-5.7	3.01
Unpaid work (domestic + care)	279.7*** (2.5)	267.0*** (2.8)	233.0*** (5.5)	264.3	18.14

Notes: Baseline: the decomposition for ages 20–39. Ages 20–26: same model on women aged 20–26. Reweighted: never-married women reweighted to the covariate distribution of married women (DiNardo-Fortin-Lemieux odds weights from a logit on single-year age, state, education, urban, caste and religion; propensities capped at 0.99), with the baseline controls. Bias-adjusted: Oster (2019) coefficient with delta = 1 and R-max = 1.3 times the controlled R-squared, relative to a regression on the married indicator alone. Breakdown delta: the delta at which the bias-adjusted coefficient is zero. FSU-clustered SEs in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Reweighting leans on a small number of never-married women who resemble the married: the effective sample of never-married women falls to 1,209 from 13,940. Trimming observations with propensities outside [0.05, 0.95] gives an unpaid-work difference of 270.5 minutes (SE 3.1) (effective never-married sample 3,006), and outside [0.10, 0.90] 268.6 minutes (SE 3.1) (3,842) (Crump et al., 2009). The learning and employment coefficients move more across these weightings (learning -70.5 and -88.7; employment -106.9 and -98.1), so the reweighted estimates for those divisions are imprecise guides. Figure 4 in the online supplement shows

the overlap of propensity scores.

Table 4 reports four checks for women. Among women aged 20 to 26, the unpaid-work coefficient is 267.0 minutes (SE 2.8), against 279.7 minutes (SE 2.5) in the baseline. After reweighting, the never-married resemble the married on observed characteristics: the largest absolute standardised difference across the covariates in Appendix Table 10 is 0.13, and Appendix Table 11 reports the coefficient-stability calculation in full. The reweighted, regression-adjusted coefficient is 233.0 minutes (SE 5.5) (without controls, 248.7 minutes (SE 5.8)). Reweighting concentrates weight on the never-married women who most resemble the married, so the effective sample of never-married women falls from 13,940 to about 1,209, and 2.0% of the weighted never-married have propensities above the cap. The bias-adjusted coefficient on unpaid work is 264.3 minutes. Selection on unobservables would have to be 18.1 times as strong as selection on observables to explain away the unpaid-work difference.

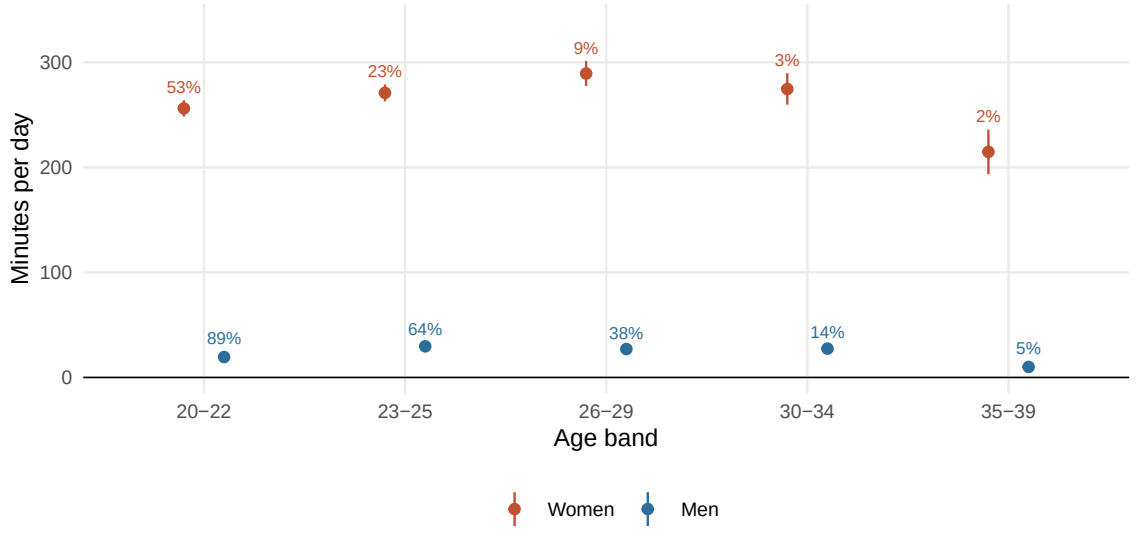


Figure 2: Married minus never-married unpaid minutes per day by age band, with 95% confidence intervals. Labels give the weighted share never married in each band.

Figure 2 shows the unpaid-work gap by age band. Among women, the coefficient is 256.10 minutes (SE 3.97) at ages 20 to 22, where 53% are never married, and 214.7 minutes (SE 10.8) at ages 35 to 39, where 2% are. The gap does not grow significantly as the never-married become rarer, which is not what strong positive selection of the never-married on unpaid-work preferences would produce. Comparisons at older ages also mix the association with marriage and the accumulation of children, which the survey links to mothers only partly.

Within married couples

Table 5: Within-couple gaps, linked couples with wives aged 20–59

	Unpaid gap (W−H)	Wife's unpaid share	Employment gap (H−W)	N
All linked couples	366.9 (1.1)	91% (0.1)	336.4 (1.5)	106,424
Wife's employment				
Wife not employed	408.1 (1.1)	91% (0.1)	411.4 (1.5)	77,619
Wife employed	257.9 (1.4)	90% (0.1)	138.3 (2.0)	28,805
Relative education				
Husband more educated	367.3 (1.4)	91% (0.1)	337.1 (2.1)	32,345
Same education tier	362.7 (1.2)	91% (0.1)	331.4 (1.7)	59,896
Wife more educated	383.0 (1.8)	91% (0.1)	355.1 (2.9)	14,183
Residence				
Head couple, no parent generation	356.5 (1.1)	91% (0.1)	328.1 (1.6)	80,850
Head couple, parent generation present	376.3 (2.3)	92% (0.2)	328.1 (3.3)	8,177
Wife is daughter-in-law of head	414.4 (1.9)	92% (0.1)	382.3 (2.4)	16,965
Spousal age gap				
Husband same age or younger	369.7 (3.9)	91% (0.3)	351.2 (6.4)	3,189
Husband older by 1-3 years	375.8 (1.5)	91% (0.1)	348.3 (2.0)	42,460
Husband older by 4-6 years	364.3 (1.4)	91% (0.1)	339.2 (2.0)	38,268
Husband older by 7-9 years	359.0 (1.9)	91% (0.1)	318.7 (2.8)	13,651
Husband older by 10+ years	345.6 (2.2)	90% (0.2)	288.1 (3.6)	8,856

Notes: Spouses linked through the household roster (head and spouse of head; married child of head and child's spouse where each is unique). Weighted by the wife's multiplier. FSU-clustered SEs in parentheses (share SE in percentage points).

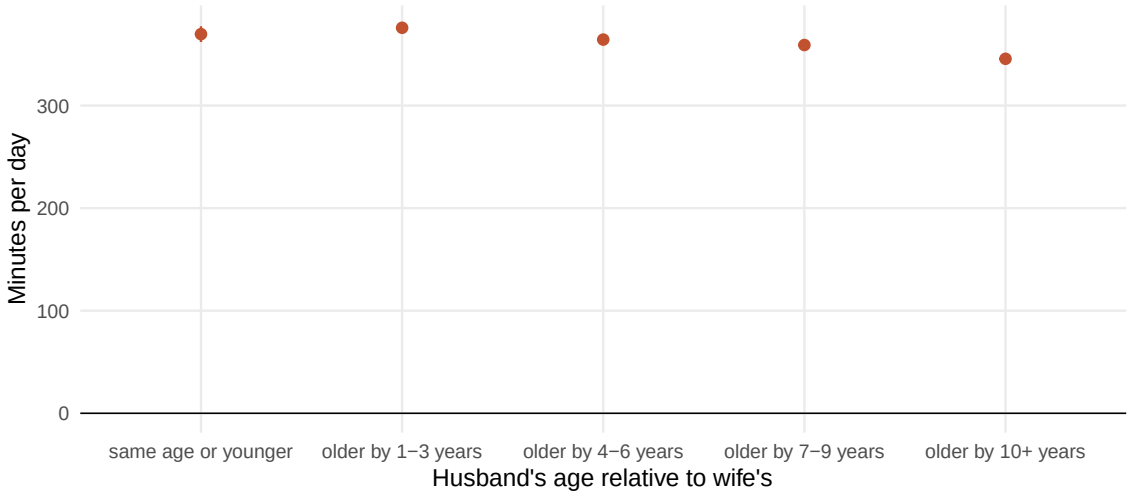


Figure 3: Wife minus husband unpaid minutes per day by spousal age gap, linked couples, with 95% confidence intervals.

Within couples, the household and the diary day are held constant. Table 5 shows that wives do 366.9 minutes (SE 1.1) more unpaid work than their husbands and account for 91% of the couple's unpaid minutes; husbands do 336.4 minutes (SE 1.5) more employment, and have 12.9 minutes (SE 0.9) more free time than their wives.

The wife–husband gap in unpaid work is 258 minutes when the wife is employed and 408 minutes when she is not, a difference of -150.2 minutes (SE 1.6). The gap in couples where the wife has more education than her husband differs from the gap in couples where the husband has more education by 15.7 minutes (SE 2.1). The gap is smaller when the wife is employed but not when she is more educated, so the evidence for H4 is mixed. Even in couples where the wife is employed, the gap remains more than half its average size. These comparisons are descriptive. Employment is chosen jointly with the division of housework, and employed wives also differ in age, education and household type.

In households where the wife is the daughter-in-law of the head, the wife–husband gap differs from that in households without a parent generation by 58.0 minutes (SE 1.9) unadjusted and 11.8 minutes (SE 1.9) after adjustment. The median husband in linked couples is 4 years older than his wife. Figure 3 plots the unpaid-work gap by the spousal age gap. Net of the wife's age, both spouses' education and location, each additional year by which the husband is older is associated with a gap that is 0.92 minutes (SE 0.22) larger, and with a wife's share that is 0.04 percentage points (SE 0.02) lower per year. Unadjusted, the gap is 370 minutes where the husband is the same age or younger and 346 where he is ten or more years older (Table 5), so in raw terms the division is less unequal when the husband is much older; large age gaps are concentrated among older wives, whose unpaid work is lower. Net of the wife's age, the minute gap is larger but the wife's share is smaller, so the evidence on the seniority interpretation of H4 is mixed. Because larger age gaps are more common when women marry young, the age gap also stands in for the wife's age at marriage and its correlates (Desai & Andrist, 2010; Field & Ambrus, 2008).

Widowhood at older ages

Table 6: Widowed minus married, minutes per day, ages 50–79

Outcome	Women	Men
Unpaid domestic and care	-73.0*** (1.7)	43.9*** (1.7)
Employment (incl. travel)	19.7*** (2.0)	-63.4*** (3.7)
Free time (socialising and leisure)	29.7*** (1.9)	12.6*** (3.0)
Sleep	25.1*** (1.1)	20.1*** (1.7)
Socialising	11.8*** (1.1)	5.0*** (1.8)
Exercise and sport	1.3*** (0.2)	1.3** (0.6)
Religious practice	2.5*** (0.6)	1.2* (0.7)

Notes: Single-year age, state and day-of-week fixed effects; controls for education, urban residence, caste, religion and proxy report. FSU-clustered SEs in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table 6 applies the same accounting to the end of marriage. Widowed women do 73.0 minutes (SE 1.7) less unpaid work than married women of the same age, and widowed men 43.9 minutes (SE 1.7) more than married men. This comparison is also selective, since widowhood is related to the spouse's health, to poverty and to the age gap between spouses, and many widowed Indians live with adult children. It shows which activities differ when a spouse is no longer present, not what widowhood causes.

Robustness

Table 7: Married minus never-married unpaid minutes per day: robustness

Specification	Women	Men
Baseline (ages 20–39)	279.7*** (2.5)	26.2*** (0.9)
Extended controls (consumption, household size, members 60+)	277.0*** (2.4)	27.0*** (0.9)
Ages 20–26	267.0*** (2.8)	26.7*** (1.4)
Ages 20–29	273.0*** (2.6)	26.8*** (1.1)
Ages 30–39	256.1*** (6.2)	22.9*** (1.5)
Self-reported diaries only	278.0*** (3.2)	23.4*** (1.3)
Urban	297.2*** (3.9)	19.8*** (1.6)
Rural	260.4*** (3.0)	30.2*** (1.2)
Excluding current students	258.0*** (2.8)	26.0*** (1.0)
Normal diary days only	287.2*** (2.6)	24.1*** (0.9)

Notes: Each row re-estimates the baseline unpaid-work coefficient on the sample or with the controls named. Extended controls add consumption quintile, household size and the number of members aged 60 and above. Normal diary days exclude days the respondent reported as not normal. FSU-clustered SEs in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table 7 re-estimates the unpaid-work coefficient for women and men under alternative samples and controls. Across specifications, the coefficient for women ranges from 256.1 to 297.2 minutes. The lowest, for ages 30 to 39 (256.14, SE 6.24), coincides at one decimal with the age 20 to 22 band coefficient (256.10, SE 3.97); the two are separate estimates on different samples. It is positive and statistically significant in every specification. Excluding current students removes the group whose time is most mechanically devoted to learning, but student status is itself affected by marriage, so this row is a bound rather than a preferred estimate.

Adding household size, older members and consumption changes the coefficient for women from 279.7 to 277.0 minutes, consistent with part of the association running through the household a woman joins at marriage.

Discussion

Because the day is fixed, more unpaid work must coincide with less of something else. This paper replaces separate estimates of marriage "penalties" in housework, sleep, social life and exercise with one decomposition in which the parts must add up. In India in 2024, married women aged 20 to 39 do substantially more unpaid domestic and care work than never-married women, and the difference is far larger for women than for men. The offsetting differences are mostly in paid work and learning, which fits the specialisation account, but specialisation in India happens against a background of low and falling married women's employment and strong norms about women's work outside the home, so it is not necessarily a choice made within the couple.

The findings speak to three debates. The first concerns how to measure the costs of marriage for women's time. Studies of single outcomes can find a sleep or leisure penalty without showing that it is the largest adjustment, or can miss that the main adjustment occurs in learning or paid work. The decomposition makes the trade-offs explicit, and it also shows where men's days change. The second concerns patrilocality. The household roster allows a direct comparison of daughters-in-law with wives who head their own household (or whose husband does). The absence of a detectable difference suggests that whatever restrictions joint households impose on young wives' autonomy (Anukriti et al., 2020; Khalil & Mookerjee,

2019) do not show up as additional unpaid minutes on average. The third concerns resources within couples. The smaller gap in couples where the wife is employed fits bargaining and relative-resources theories, but the within-couple gap remains large across all the groups in Table 5, which is consistent with arguments that gender works at the institutional and interactional levels as well as through the resources of individuals (Ridgeway & Correll, 2004; Risman, 2004).

Limitations

The main limitation is the comparison group. Married and never-married women differ in ways that age, education and location do not capture, and the survey does not record when a marriage began. The four selection checks narrow the range of plausible interpretations but do not identify the causal effect of marriage. The estimates are best read as the time-use profile that accompanies marriage in contemporary India, with the restricted-age and reweighted estimates as the most credible magnitudes. Second, the couple analysis covers only spouses whom the roster can link without ambiguity. Joint households with several married sons are excluded, and these are probably the households in which hierarchy among daughters-in-law matters most. Third, the residence categories describe relationships to the household head, not to the woman's own parents-in-law, and the "parent generation present" category mixes the husband's and the wife's parents. Fourth, a single diary day per person measures daily time use with error. This adds noise but should not bias the comparisons of means, except where the diary day differs systematically by marital status; the day-of-week effects and the normal-day check address the most obvious source. Fifth, I do not condition on children.

Parenthood is a principal channel through which marriage changes time use (Baxter et al., 2008; Sanchez & Thomson, 1997), and conditioning on it would answer a different question. Panel or retrospective data on the timing of marriage and births would allow an event-study design in the style of Kleven et al. (2019) that separates the two transitions.

Conclusion

Married and never-married Indian women spend the day very differently. Married women do far more unpaid domestic and care work, a difference with no counterpart among men. A time-budget decomposition shows which activities differ alongside that work, and the household roster shows how the change depends on whether a young wife joins her husband's parents' household and on her standing relative to her husband. Policies that aim to raise married women's employment, such as childcare, safe transport or flexible work, operate within this budget. They are more likely to succeed if they also reduce the unpaid domestic and care work that accompanies marriage and motherhood, since that work competes with women's paid work, learning, leisure or sleep.

Declaration of generative AI use

Generative AI tools (Claude, Anthropic) were used to assist with writing and debugging analysis code, checking references, and drafting and editing text. The author reviewed and verified all analyses, results, and text, and takes full responsibility for the content.

Data and code availability

The Time Use Survey 2024 unit-level data are distributed by the National Statistics Office, Ministry of Statistics and Programme Implementation, Government of India (microdata.gov.in, catalogue 236), to registered users. Code available from the author on request; will be made public on acceptance.

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Appendix

Table 8: From the household roster to the analysed diaries

Step	N
Persons aged 6+ on the household roster	454,192
Without diary records in the file	3,735
Diaries (person-days)	450,457
Diaries not summing to exactly 1,440 minutes	5
Third-gender diaries (excluded from women-men comparisons)	49
Diaries of women and men	450,408

Notes: All counts are computed from the microdata at knit time. Persons without diary records appear only on the household roster.

Table 9: Activity categories and ICATUS 2016 codes

Category	ICATUS 2016 codes
Employment	110, 121–129, 131–136, 139, 141–142, 150, 160, 170
Employment-related travel	181–182
Own-use production of goods	211–218, 221–227, 229–230, 241–242, 250
Food and meals	311–314, 319
Cleaning and upkeep	321–325, 329
Care of textiles	341–344, 349
Shopping	371–372
Unpaid domestic services (division 3)	311–314, 319, 321–325, 329, 331–333, 339, 341–344, 349, 351–352, 359, 361–362, 369, 371–372, 380, 390
Childcare	411–417, 419, 442
Care of dependent adults	421–426, 429, 443
Unpaid caregiving (division 4)	411–417, 419, 421–426, 429, 431–432, 439, 441–444, 490
Learning	611–614, 619–620, 630, 640, 690
Free time (divisions 7 and 8)	711–713, 719, 721–723, 729–730, 741–742, 749–750, 790, 811–813, 819, 821–823, 829, 831–832, 841–843, 849–850, 860, 890
Sleep	911–913, 919

Notes: Minutes are summed over major activities in the person-day diary. Codes are three-digit ICATUS 2016 codes as recorded in the survey.

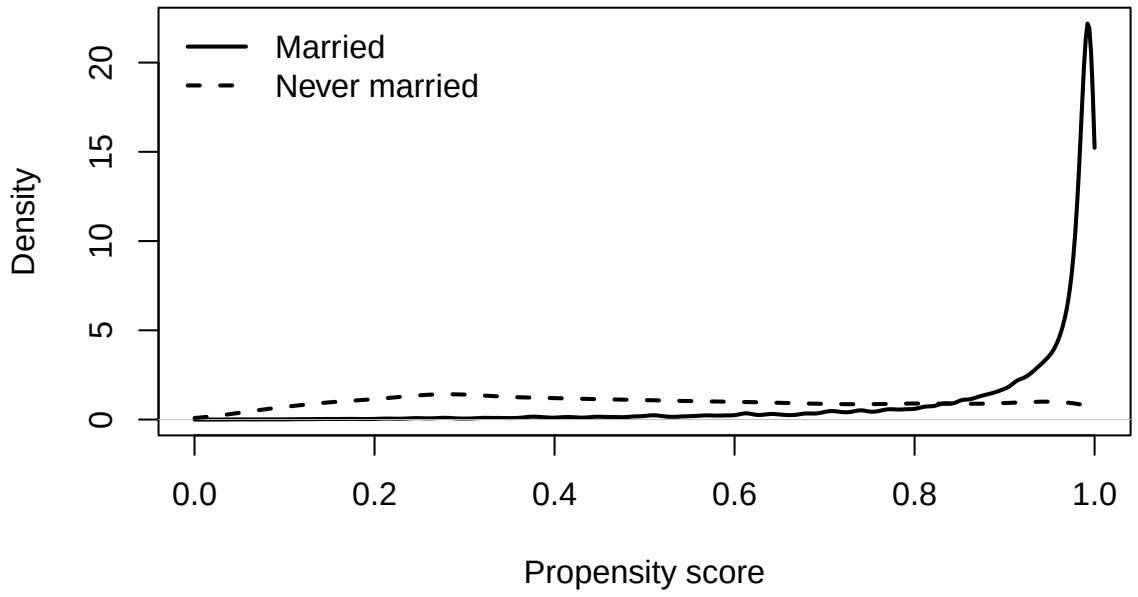


Figure 4: Estimated propensity of being married, women aged 20–39, by marital status (survey-weighted densities)

Table 10: Covariate balance before and after reweighting, women aged 20–39

Covariate	Married	Never married	Never married, reweighted	Std. diff. before	Std. diff. after
Age (years)	29.84	23.39	29.79	1.46	0.01
Secondary education or more	0.48	0.85	0.41	-0.85	0.13
Graduate	0.16	0.45	0.14	-0.68	0.05
Urban	0.29	0.43	0.31	-0.30	-0.03
Scheduled Caste or Tribe	0.30	0.28	0.34	0.04	-0.10
Muslim	0.14	0.12	0.16	0.07	-0.06

Notes: Weighted means. Standardised difference: difference in means divided by the square root of the average of the two group variances. Reweighting uses DiNardo-Fortin-Lemieux odds weights (see the selection table in the main text).

Table 11: Coefficient stability for women: married minus never-married, minutes per day

Outcome	Uncontrolled	Controlled	Bias-adjusted	Breakdown delta
Unpaid domestic and care work	290.5	279.7	264.3	18.14
Employment (incl. travel)	-50.3	-84.7	-96.2	-7.40
Learning	-152.3	-118.7	-43.0	1.57
Socialising, community, religion	-1.5	-6.6	-8.2	-4.30
Culture, leisure, media, sport	-56.9	-41.3	-33.1	5.04
Sleep	-17.1	-8.6	-5.7	3.01

Notes: Uncontrolled: regression on the married indicator only. Controlled: baseline model. Bias-adjusted and breakdown delta as in the selection table (Oster 2019, R-max = 1.3 times the controlled R-squared).

Table 12: Construction and validation of the person-day panel

Check	Value
Activity records	10,211,478
Person-days (diaries)	450,457
Households	139,487
First Stage Units	9,969
Diaries summing to 1,440 minutes	100.0%
Records flagged minor (secondary)	17.5%
Persons with varying weight	0.00%
Invalid activity codes	None
Married persons with linked spouse	88.5%
Under-6 count method	direct count (file contains members under 6)

Notes: All activity records are used; no row sampling.